

Input:

- V . Set of nodes
- E_s . Set of scheduled edges $(v, u, \text{dep}, \text{arr})$
- E_t . Set of transfer edges (v, w, Δ)
- $\text{country}(v)$. Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- $MTT(c)$. Precomputed minimum transit times for each country
- $\text{bucket}(t)$. Time bucket mapping function
- record . Current record to beat (e.g., 13). Static.
- rollout_freq . Number of iterations between rollouts (e.g., 1000)
- GlobalMinTime . Minimum realistic time to visit a country (e.g., 30 mins)

Output:

- P . Sequence of labels representing a route within 24 hours

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1: Initialize  $\text{Labels}(v) \leftarrow \emptyset$  for all  $v \in V$ 
2: Initialize priority queue  $Q \leftarrow \emptyset$  // ordered by decreasing score
3: Initialize best label  $\leftarrow \text{null}$ 
4:  $\text{iters} \leftarrow 0$  // Iteration counter for rollouts
5: // Initialize labels at departures
6: for all  $v \in V$  do
7:   for all scheduled edges  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
8:      $L \leftarrow (u, \text{arr}, \{\text{country}(u)\}, \text{pred} = \text{null}, \text{start\_time} = \text{dep})$ 
9:      $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L\}$ 
10:    compute score( $L$ )
11:     $Q \leftarrow Q \cup \{L\}$ 
12: // Main loop
13: while  $Q \neq \emptyset$  do
14:   extract  $L = (v, t, C, \text{pred}, \text{start\_time})$  with maximum score from  $Q$ 
15:   // — GLOBAL HARD FLOOR PRUNING —
16:    $\text{needed} \leftarrow (\text{record} + 1) - |C|$ 
17:    $\text{remaining\_time} \leftarrow 1440 - (t - \text{start\_time})$ 
18:   // If we need countries but lack the physical time (e.g., j 30 mins per country), kill it.
19:   if  $\text{needed} > 0 \wedge \text{remaining\_time} < (\text{needed} \times \text{GlobalMinTime})$  then
20:     continue
21:   // —————
22:   // — MONTE CARLO ROLLOUT TRIGGER —
23:    $\text{iters} \leftarrow \text{iters} + 1$ 
24:   if  $\text{iters} \bmod \text{rollout\_freq} = 0$  then
25:     // Unconditional rollout for any label that survived Hard Floor
26:      $L_{\text{rollout}} \leftarrow \text{Algorithm 7}(L, E_s, E_t, \text{country})$ 
27:     if  $\text{best label} = \text{null} \vee |L_{\text{rollout}}.C| > |\text{best label}.C|$  then
28:        $\text{best label} \leftarrow L_{\text{rollout}}$ 
29:       // Early stopping if rollout beats the static record
30:       if  $|\text{best label}.C| > \text{record}$  then
31:         break
32:   // —————

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33: // Enforce 24-hour journey limit (1440 absolute minutes)
34: if remaining_time < 0 then
35:   continue
36: // Upper bound computation (Time-Budgeted)
37:  $b \leftarrow \text{bucket}(t)$ 
38:  $U \leftarrow R(v, b) \setminus C$ 
39: Sort  $U$  as a sequence  $(c_1, c_2, \dots, c_k)$  such that  $MTT(c_1) \leq MTT(c_2) \leq \dots \leq MTT(c_k)$ 
40: time_spent  $\leftarrow$  0
41: UB  $\leftarrow$  0
42: for  $i = 1$  to  $|U|$  do
43:   time_spent  $\leftarrow$  time_spent +  $MTT(c_i)$ 
44:   if time_spent  $\leq$  remaining_time then
45:     UB  $\leftarrow$  UB + 1
46:   else
47:     break
48: // Record-based pruning (Using static record)
49: if  $|C| + \text{UB} \leq \text{record}$  then
50:   continue
51: // Update best label from normal queue expansion
52: if best_label = null  $\vee |C| > |\text{best\_label}.C|$  then
53:   best_label  $\leftarrow$   $L$ 
54: // Early stopping if normal expansion beats the static record
55: if  $|C| > \text{record}$  then
56:   break
57: // Extend via scheduled edges
58: for all  $(v, u, \text{dep}, \text{arr}) \in E_s$  do
59:   if  $\text{dep} \geq t$  &  $\text{dep} - \text{start\_time} \leq 1440$  then
60:      $t_{\text{new}} \leftarrow \text{arr}$ 
61:     if  $t_{\text{new}} - \text{start\_time} > 1440$  then
62:       continue
63:      $C' \leftarrow C \cup \{\text{country}(u)\}$ 
64:      $L' \leftarrow (u, t_{\text{new}}, C', \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
65:     if  $\nexists L'' \in \text{Labels}(u)$  such that
66:        $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C' \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C'))$  then
67:         remove all  $L'' \in \text{Labels}(u)$  such that
68:            $(t_{\text{new}} \leq L''.t \wedge C' \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C' \neq L''.C))$ 
69:          $\text{Labels}(u) \leftarrow \text{Labels}(u) \cup \{L'\}$ 
70:         compute score( $L'$ )
71:          $Q \leftarrow Q \cup \{L'\}$ 
72: // Extend via transfers
73: for all  $(v, w, \Delta) \in E_t$  do
74:    $t_{\text{new}} \leftarrow t + \Delta$ 
75:   if  $t_{\text{new}} - \text{start\_time} > 1440$  then
76:     continue
77:    $L' \leftarrow (w, t_{\text{new}}, C, \text{pred} = L, \text{start\_time} = \text{start\_time})$ 
78:   if  $\nexists L'' \in \text{Labels}(w)$  such that
79:      $(L''.t \leq t_{\text{new}} \wedge L''.C \supseteq C \wedge (L''.t < t_{\text{new}} \vee L''.C \neq C))$  then
80:       remove all  $L'' \in \text{Labels}(w)$  such that
81:          $(t_{\text{new}} \leq L''.t \wedge C \supseteq L''.C \wedge (t_{\text{new}} < L''.t \vee C \neq L''.C))$ 
82:        $\text{Labels}(w) \leftarrow \text{Labels}(w) \cup \{L'\}$ 

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79:   |   |   |   compute score( $L'$ )
80: |   |   |    $Q \leftarrow Q \cup \{L'\}$ 

81: // Reconstruct path
82:  $P \leftarrow$  empty sequence
83:  $L \leftarrow$  best label
84: while  $L \neq$  null do
85: |   prepend  $L$  to  $P$ 
86: |    $L \leftarrow L.\text{pred}$ 
87: return  $P$ 

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